

EN- Based CONCRETE Codes – Currently in Singapore

SS EN 1992-1-1 : 2008

Eurocode 2 : Design of concrete structures

Part 1-1 : General rules and rules for buildings

Concrete : characteristic compressive cylinder strength adopted in design equations, with equivalent characteristic cube compressive strength for production conformity, specified in National Guidance Document (SS 544)

SS EN 206-1 : 2009, Specification for Concrete

Part 1: Specification, performance, production and conformity

Complementary Singapore Standards

SS 544: 2009 Concrete - Complementary Singapore Standard to SS EN 206-1

Part 1 : Method of specifying concrete and guidance for the specifier

Part 2 : Specification for constituent materials and concrete

Currently under revision to update based on:

BS EN 206: 2013 (SS EN 206-1: 2009)

BS 8500-1: 2006+A1: 2012 (SS 544-1: 2009)

BS 8500-2: 2006+A1: 2012 (SS 544-2: 2009)

EN STANDARDS FOR CONCRETE

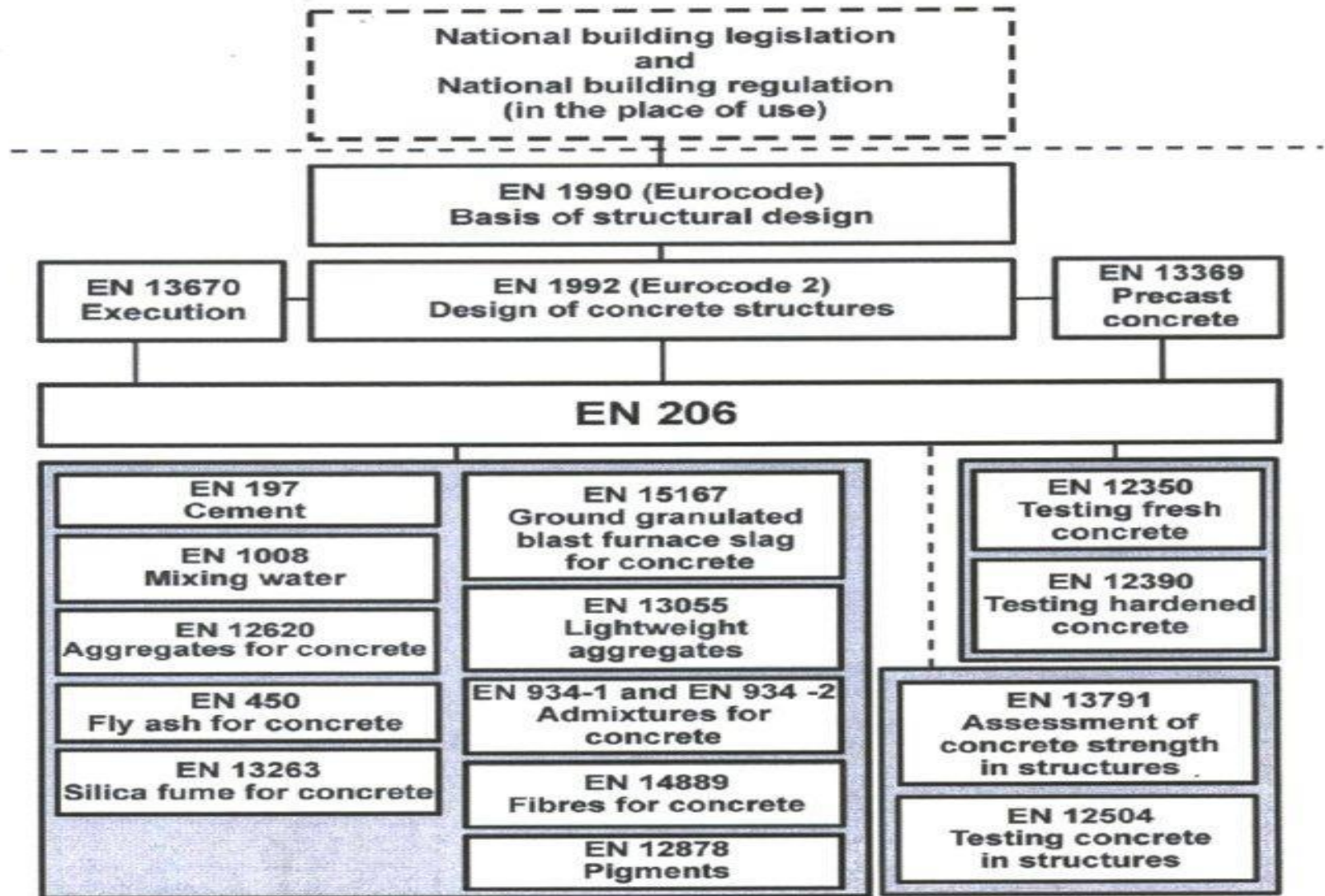


Figure 1 — Relationships between EN 206 and standards for design and execution, standards for constituents and test standards

Specification of Concrete (SS EN-206-1: 2009)

SS 544-1: 2009, Concrete – Complementary Singapore Standards to SS EN 206-1, Part 1: Method of specifying and guidance for the specifier

(Based on BS 8500-1: 2006, currently under revision to update to BS 8500-1: 2006+A1:2012)

Annex A (informative), Guidance for the specifier

More specific recommendations for resistance to identified exposure classes than in SS EN 206-1

Ground conditions – BRE Special Digest 1: 2005

Annex B (normative), Identity testing for slump, flow, air content, density and additional requirements for compressive strength

For alkali-silica reaction guidance from BRE Digest 330 is replaced by Annex D in BS 8500-2:2006+A1: 2012

Specification of Concrete (SS EN-206-1: 2009)

SS 544-2: 2009, Concrete – Complementary Singapore Standards to SS EN 206-1, Part 2: Specification for constituent materials and concrete

(Based on BS 8500-2: 2006, currently under revision to update to BS 8500-2: 2006+A1:2012)

Annex A (normative), Conformity procedure for combinations

Annex B (normative), Test method for determining the composition of RCA and RA

Annex D (normative), Minimizing the risk of damaging alkali-silica reaction in concrete

(Added to BS 8500-2:2006+A1: 2012 to replace guidance from BRE Digest 330 is in BS 8500-2:2006/SS 544-2: 2009)

Specification of Concrete

SPECIFYING CONCRETE

The following topics relating to specification of concrete are considered, using the case of designed concrete to illustrate the approach:

- (a) the types of concrete
- (b) general requirements for design concrete
- (c) additional requirements for design concrete
- (d) strength classes, consistence classes and exposure classes
- (e) conformity test and requirements for compressive strength
- (f) initial test and identity test

Where appropriate, the difference between the new approach and the approach based on SS 289 (BS 5328) (withdrawn) is highlighted.

Constituent materials for concrete

Types of cement, aggregate, water and admixtures
(see other presentations)

Specification of Concrete

EN 206-1 Definitions

3.1.1 Concrete

Material formed by mixing cement, coarse and fine aggregate and water, with or without incorporation of admixtures and additions, which develops its properties by hydration of the cement

3.1.5 Ready-mixed concrete

Concrete delivered in a fresh state by a person or body who is not the user. Ready-mixed concrete in the sense of this standard is also:

- concrete produced off site by the user**
- concrete produced on site, but not by the user**

3.1.39 User

Person or body using fresh concrete in the execution of a construction or a component

3.1.38 Producer

Person or body producing fresh concrete

3.1.37 Specifier

Person or body establishing the specification for the fresh and hardened concrete

Specification of Concrete

EN 206-1: Definitions

3.1.32 Characteristic strength

Value of strength below which 5% of the population of all possible strength determination of the volume of concrete under consideration, are expected to fall

3.1.41 Initial test

Test or tests to check before production starts how a new concrete or concrete family shall be composed in order to meet all the specified requirements in the fresh and hardened concrete

3.1.42 Identity test

Test to determine whether selected batches or loads come from a conforming population

3.1.43 Conformity test

Test performed by the producer to assess conformity of the concrete

3.1.14 Concrete family

Group or concrete compositions for which a reliable relationship between relevant properties is established and documented

Specification of Concrete

Types of Concrete

EN 206 Definitions

3.1.11 Designed concrete

Concrete for which the required properties and additional characteristic are specified to the producer who is responsible for providing a concrete conforming to the required properties and additional characteristics

3.1.12 Prescribed concrete

Concrete for which the composition of the concrete and the constituent materials to be used are specified to the producer who is responsible for providing a concrete with the specified composition

3.1.13 Standardized prescribed concrete

Prescribed concrete for which the composition is given in valid in the place of use of the concrete

NOTE: WORDS SHOWN IN ITALICS AND UNDERLINED FOR EMPHASIS

Specification of Concrete

Types of Concrete

SS 544 - 1 (BS 8500):

3.1.5 Designated concrete

Concrete produced in accordance with SS 544-2 : 2009, by a producer holding current accredited product conformity certification based on product testing and surveillance, coupled with approval of the producer's quality system to BS ISO 9001.

SS 544-1 National Foreword highlighted to local users:

Designated concrete is derived based on UK experience and materials. Local users are advised to make appropriate judgement on the relevance/suitability of the designated concrete for the intended use in the specific environmental condition.

3.1.10 Proprietary concrete

Concrete for which the producer assures the performance subject to good practice in placing, compacting and curing and for which the producer is not required to declare the composition

NOTE: WORDS SHOWN IN ITALICS AND UNDERLINED FOR EMPHASIS

Specification of Concrete

SS 544 - 1 (BS 8500) Definitions

3.1.2 Combination

Restricted range of Portland cements and additions which, having been combined in the concrete mixer, count fully towards the cement content and water/cement ratio in concrete

3.1.4 Design chemical class (DC-class)

Designation used to describe a concrete quality capable of resisting the selected aggressive chemical environment for the concrete, provided that any specified additional protective measures (APMs) are correctly applied to the structure

3.1.11 Recycled aggregate (RA)

Aggregate resulting from the reprocessing of inorganic materials previously used in construction

3.1.12 Recycled concrete aggregate (RCA)

Recycled aggregate principally comprising crushed concrete

3.1.9 nominal cover

Depth of cover to reinforcement shown on the drawings comprising the minimum cover plus an allowance in design for deviation, Δc , to accommodate fixing precision

Specification of Concrete

BS 8500-1: 2006+A1: 2013 Definitions

3.1.13 Alkali content of concrete

Value calculated from the mix proportions and the determined alkali contents of each of the constituents and used for verifying that the alkali content of concrete does not exceed the specified limit

3.1.14 Certified average alkali content

Average of 25 consecutive determinations of alkali content carried out on spot samples and expressed as sodium oxide equivalent

3.1.15 Declared mean alkali content

Alkali content of cement or addition, expressed as the sodium oxide equivalent (Na_2O eq.) as declared by the cement or addition manufacturer

3.1.16 Guaranteed alkali limit

Alkali limit, expressed as sodium oxide equivalent (Na_2O eq.) which the constituent material supplier guarantees is not exceeded by any test result on any spot sample

3.1.18 Sodium oxide equivalent (Na_2O eq.)

Alkali content, as a percentage, calculated as:

$$\% \text{Na}_2\text{O eq.} = \% \text{Na}_2\text{O} + 0.658\% \text{K}_2\text{O}$$

Specification of Concrete

EN 206 Clause 6 Specification of concrete

The specifier of the concrete shall ensure that all the relevant requirements for concrete properties are included in the specification given to the producer.

(designer & contractor to provide)

The specifier shall also specify any requirements for concrete properties that are needed for transportation after delivery, placing, compaction, curing or further treatment.

(contractor in particular)

The specification shall, if necessary, include any special requirements (e.g. for obtaining an architectural finish).

(designer & contractor to provide)

Recommendation:

Performance requirements, where possible, shall be specified in quantitative term, within a range or a target value with tolerance and test method to assess conformity (by mutual agreement between specifier, user and producer where no standard test method is available)

Specification of Concrete

EN 206 Clause 6 Specification of concrete

The specifier shall take account of:

- the application of the fresh and hardened concrete;
- the curing conditions;
- the dimensions of the structure (the heat development);
- the environmental actions to which the structure is to be exposed;
- any requirements for exposed aggregate or tooled concrete finishes;
- any requirements related to the cover to reinforcement or minimum section width, e.g. maximum nominal upper aggregate size;
- any restrictions on the use of constituent materials with established suitability, e.g. resulting from exposure classes

Comment:

Curing method and duration have greater influence on concrete quality in the cover zone, affecting durability performance, than for the bulk

Environmental actions may place restrictions on specific types of cement suitable for use with specified minimum cover and selected strength class (or maximum w/c + minimum cement content)

Specification of Concrete

- **For designed concrete (performance)**
 - Basic requirements (normal weight concrete):
 - Compressive strength class
 - Exposure class
 - Maximum nominal upper aggregate size
 - Chloride content class
 - Consistence class or target value of consistence
 - Additional requirements:
 - Special types or classes of cement, e.g. low heat
 - Requirements for temperature of fresh concrete
 - Strength and heat development
 - Resistance to water penetration or abrasion
- **SS EN 206-1 Annex J (informative) – performance-related design methods with respect to durability (SS 544-1: 2009 provision for proprietary concrete)**

Specification of Concrete

Specification for Designed Concrete – Example to illustrate approach
(SS 544-1 or BS 8500-1 in lieu of SS EN 206-1 to provide more details)

4.3.3 Additional requirements

(selected items only – refer to Clause 4.3.3 for full list)

- (a) special types or classes of aggregate, e.g. for wear resistance or freeze-thaw resistance (see Note 1)

Note 1 – In these cases, EN 206-1, 6.2.3 suggests that concrete composition to minimize damaging alkali-silica reaction is the responsibility of the specifier. However, BS 8500-2 (SS 544-2), 5.2 requires the producer to take appropriate action in these cases, (see A.8.1 for explanation).

A.8.1 Alkali-aggregate reaction

“If the producer follows the guidance given in BRE Digest 330. this will be deemed to have satisfied the requirement to minimize damaging alkali-silica reactions”.

BRE Digest 330, Alkali-silica reaction in concrete, (in four Parts), Building Research Establishment, 2004.

Replaced in BS 8500-2:2006+A1: 2013 with Annex D (normative)

Specification of Concrete

Specification for Designed Concrete – Example to illustrate approach
(SS 544-1 or BS 8500-1 in lieu of SS EN 206-1 to provide more details)

4.3.3 Additional requirements (continued)

(selected items only – refer to Clause 4.3.3 for full list)

- (f) requirements for the temperature of the fresh concrete, where different from the lower limit in EN 206-1, 5.2.8 **(not less than 5°C)** or the upper limit in SS 544-2, 5.4 **(35°C unless specified)**

Comments:

Fresh concrete temperature together with temperature rise from heat development during hydration of cement lead to maximum temperature developed at the interior of thick concrete sections. The limiting maximum temperature is to minimize potential delayed ettringite formation (DEF).

For CEM I, DEF may occur at temperatures of 65° to 70°C and for other types of cement, e.g. CEM III/B a higher temperature may be permitted up to 75°C

Ref: Quillin, K, Delayed ettringite formation – In-situ concrete, Information Paper IP11/01, Building Research Establishment, 2001

Specification of Concrete

Specification for Designed Concrete – Example to illustrate approach
(BS 8500-1 & SS 544-1 – in lieu of EN 206-1 for more details)

4.3.3 Additional requirements (continued)

(selected items only – refer to Clause 4.3.3 for full list)

(g) strength development (see Table 12 in EN 206-1: $f_{cm,2}/f_{cm,28}$)

Comment:

For the determination of curing time, information on the strength development of the concrete may be given either in terms of Table 12 or by a strength development curve at 27 °C between 2 and 28 days (from initial tests)

Table 12 – Strength development of concrete at 27 °C

Strength development	Estimate of strength ratio, $f_{cm,2}/f_{cm,28}$
Rapid	≥ 0.5
Medium	≥ 0.3 to < 0.5
Slow	≥ 0.15 to < 0.3
Very slow	< 0.15

Specification of Concrete

Specification for Designed Concrete – Example to illustrate approach
(BS 8500-1 & SS 544-1 – in lieu of EN 206-1 for more details)

4.3.3 Additional requirements (continued)

(selected items only – refer to Clause 4.3.3 for full list)

(h) heat development during hydration

Comment:

Strength development and heat development go hand in hand and the use of low heat cement will lead to low early strength which user has to be taken into consideration when specifying such concrete, e.g. CEM III/B 32.5-LH with characteristic value for heat of hydration ≤ 270 J/g in accordance with either EN 196-8 at 7 days or in accordance with EN 196-9 at 41 h. Upper limit value for single results – 300 J/g

Characteristic value – from quality control testing by cement producer over the most recent assessment period (not from single test sample)

Pre-blended cement characteristic value can be obtained from producer, but not when using combination (single test result only for specific sample)

Specification of Concrete

**Specification for Designed Concrete – Example to illustrate approach
(BS 8500-1 & SS 544-1 – in lieu of EN 206-1 for more details)**

4.3.3 Additional requirements (continued) (selected items only – refer to Clause 4.3.3 for full list)

(i) retarded stiffening

Comment:

Delayed stiffening is needed to avoid the potential for formation of a cold joint between concrete recently placed and new fresh concrete placed over it to achieve homogeneity.

The required period of retardation is achieved by the addition of retarding admixture at appropriate dosage determined during initial test (trial mixes) based on the “initial setting time of concrete” in accordance with ASTM C 403 or equivalent test method (SS 320, BS 5075)

EN 934 Admixtures for concrete, mortar and grout, has adopted a test method based on standard mortar using the “Vicat” needle for the determination of delay in setting.

Specification of Concrete

Specification for Designed Concrete – Example to illustrate approach
(SS 544-1 or BS 8500-1 in lieu of SS EN 206-1 to provide more details)

4.3.3 Additional requirements (continued)

(selected items only – refer to Clause 4.3.3 for full list)

(j) resistance to water penetration

Comment:

This is also known as waterproofed concrete. Penetration resistance is linked to a particular pressure head. Direct method of assessment is difficult to perform as it may need to apply a long sustained pressure. Indirect indication may be obtained by determining water penetration depth under specific test conditions, e.g. EN 12390-8: 2000 Testing hardened concrete – Part 8: Depth of penetration of water under pressure. However, the acceptability criteria for a given project have to be stated by specifier in relation to applied water pressure of (500 ± 50) kPa for (72 ± 2) h adopted in the test.

(l) tensile splitting strength

Concrete for pavement construction

Specification of Concrete

- **For prescribed concrete (not commonly specified)**
 - Basic requirements:
 - Cement content
 - Cement type and strength class
 - w/c ratio or consistence in terms of class or a target value
 - Consistence class or target value consistence
 - Type, category and max chloride content of aggregate
 - Maximum aggregate size and limitation for grading
 - Type and quantity of admixture or addition, if any
 - Additional requirements:
 - Sources of concrete constituents
 - Requirements for the temperature of fresh concrete
 - Other technical requirements
- ★ **Producer responsible for providing concrete with the specified composition only (specifier responsible for performance)**

STRENGTH CLASSES

**(Compressive strength – as example)
(see EN 206 for tensile strength)**

Specification of Concrete

EN 206-1 Clause 4.3.1

Compressive strength classes

The characteristic compressive strength at 28 days of 150 mm diameter by 300 mm cylinder ($f_{ck,cyl}$) or the characteristic compressive strength at 28 days of 150 mm cubes ($f_{ck,cube}$) may be used for classification.

SS 544-2 (BS 8500-2) Clause 12.2 Conformity for compressive strength accepts the characteristic compressive strength at 28 days of 100 mm cubes to be treated in the same way for acceptance as for 150 mm cubes.

(Ratio of test specimen size to maximum aggregate size of at least 3 is adopted in ASTM standards and for diameter of cores in EN 13791: 2007, British standards (BS 1881), a ratio of 4 was adopted. In the case of 20 mm maximum aggregate size, 100 mm is already 5 times)

SS EN 206-1: 2009 (BS EN 206-1) Table 7 – Compressive strength classes for normal-weight and heavy-weight concrete

SS EN 206-1 (BS EN 206-1) Table 8 – Compressive strength classes for light-weight concrete

SS 544-2 (BS 8500-2) Table 9 – Additional compressive strength classes to those given in SS EN 206-1 (BS EN 206-1)

Specification of Concrete

Table 7 – Compressive strength classes for normal-weight and heavy-weight concrete
(factor of 1.25 up to C55/67, then difference of 15 N/mm²)

Compressive strength class	Minimum characteristic cylinder strength, $f_{ck,cyl}$ (N/mm ²)	Minimum characteristic cube strength $f_{ck,cube}$ (N/mm ²)
C8/10	8	10
C12/15	12	15
C16/20	16	20
C20/25	20	25
C25/30	25	30
C30/37	30	37
C35/45	35	45
C40/50	40	50
C45/55	45	55
C50/60	50	60
C55/67	55	67
C60/75	60	75
C70/85	70	85
C80/95	80	95
C90/105	90	105
C100/115	100	115

Specification of Concrete

BS 8500-2, (SS 544-2)

Table 9 – Additional compressive strength classes to those given in EN 206-1

Compressive strength class	Minimum characteristic cylinder strength, $f_{ck,cyl}$ (N/mm ²)	Minimum characteristic cube strength, $f_{ck,cube}$ (N/mm ²)
C6/8	6	8
C28/35	28	35
C32/40	32	40
LC28/31	28	31
LC32/35	32	35

Comment:

In Table 7 and Table 9, up to C55/67, $f_{ck,cube} \cong 1.25 \times f_{ck,cyl}$

From C60/75 onwards, $f_{ck,cube} = f_{ck,cyl} + 15$ (N/mm²) [$f_{ck,cube} / f_{ck,cyl}$ decreasing with f_{ck}]

Recommend to have both cylinder and cube compressive strength tested in initial tests (trial mix) to provide confidence in using Table 7 and Table 9 between structural design based on cylinder strength and production conformity based on cube strength.

EN 2060: 150 mm diameter by 300 mm cylinders or 150 mm cubes

SS 544-2 (BS 8500-2), Clause 12.2: when using 100 mm cubes both Table 7 of

EN 206-1 and Table 9 of SS 544-2 (BS 8500-2) for 150 mm cubes remain valid

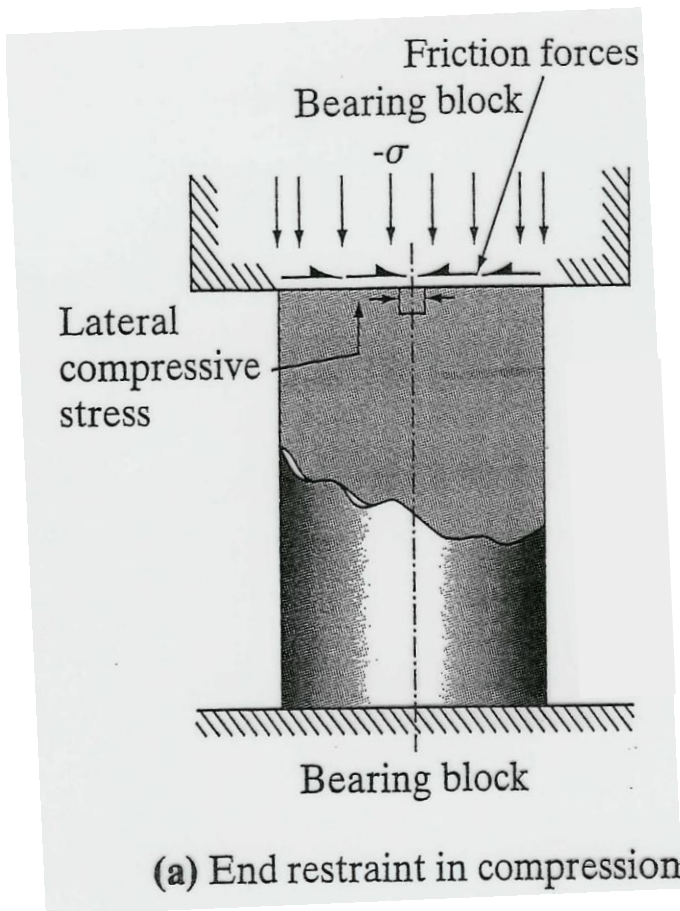
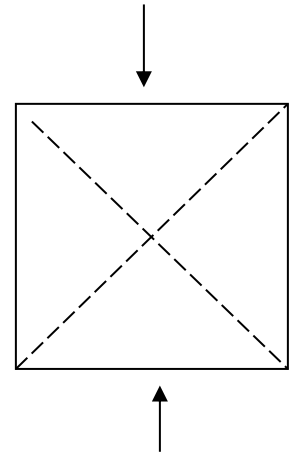
Recommend: During transition to 100 mm cube size, both sizes should be tested during initial testing and early stage of project to provide confidence.

Specimen shape and size

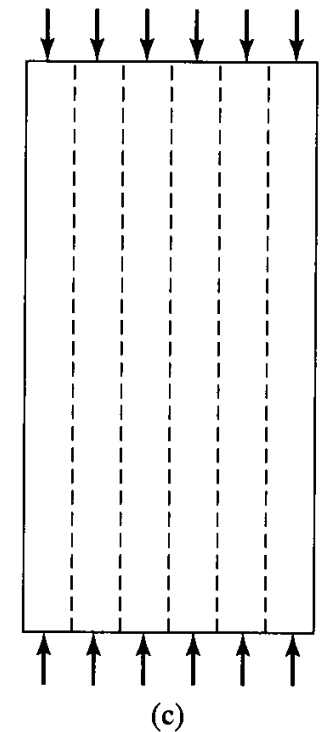
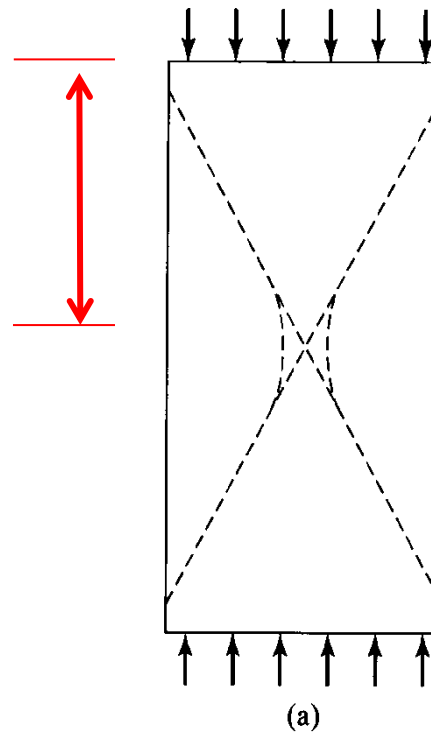
ASTM: cylinder specimens (100x200 or 150x300 mm)

EN and SS: cubes (100 or 150 mm)/cylinder (150x300 mm)

- **End restraint and failure mode**



0.866 d



Influence of End Restraint

Material	Modulus of elasticity (GPa)	Poisson's ratio
Concrete	20 to 40	0.10 to 0.20
Steel	180 to 200	0.27 to 0.30
Steel/concrete	4.5 to 10	1.4 to 3.0

Under same uniaxial stress, for low end of concrete strength, axial strain in concrete can be up to 10 times that of steel (200/20) but much lower, around 5 (200/40) for high strength concrete

Assuming ratio of Poisson's of steel/concrete = 2 (0.30/0.15)

For low end of concrete strength, lateral concrete strain is then about 5 times that of steel and reduced to around 2.5 times at the high end of concrete strength

The lateral expansion strain of steel platen is much less than that of end of concrete specimen, (if free of friction between them).

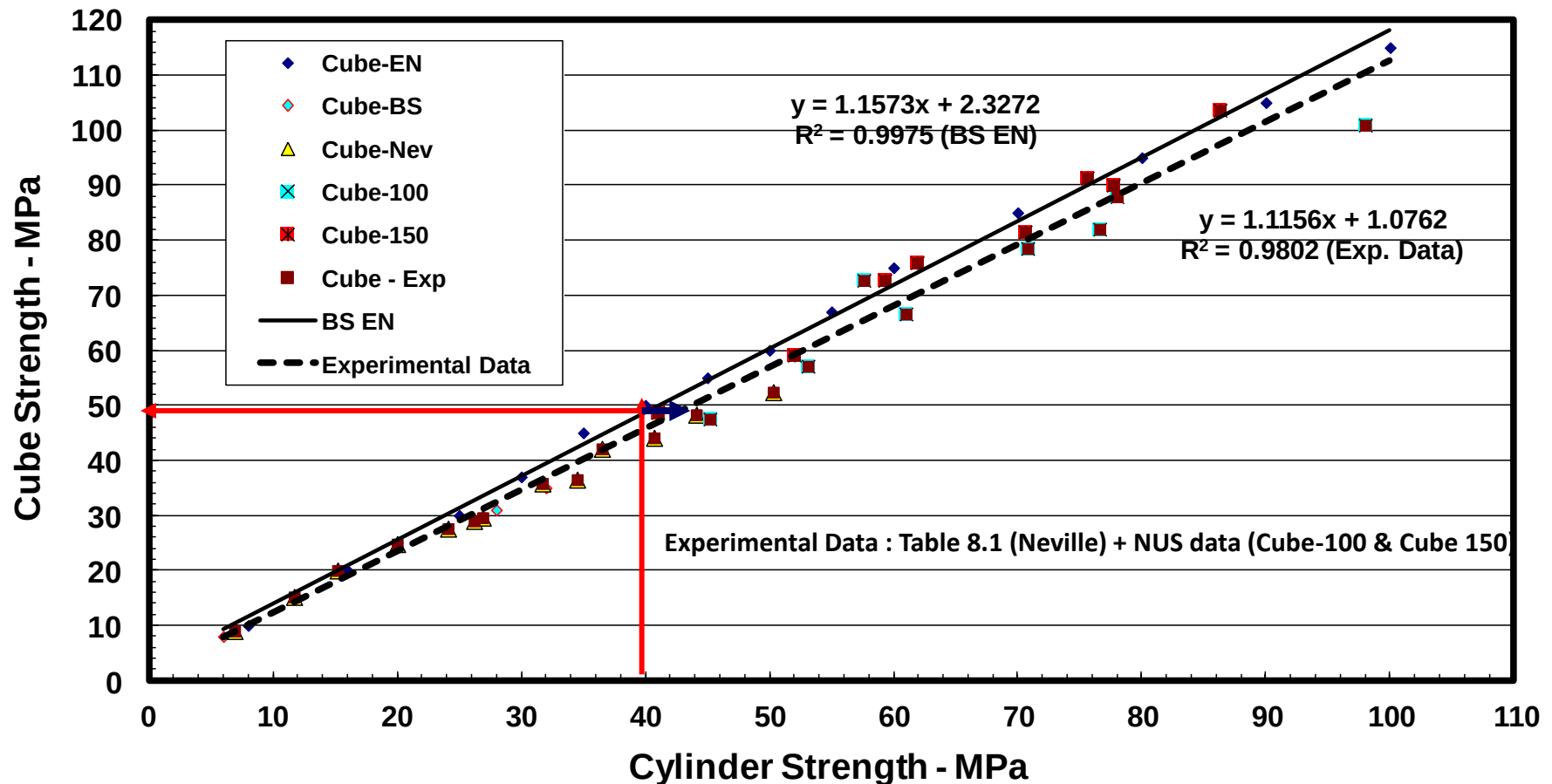
End friction induced confining stress, highest at interface and diminishing with distance towards the centre of specimen ($\sqrt{3}/2 \times d$) where d = diameter of specimen, but expected to be less with lower ratio of difference between free lateral strain).

Based on limited experimental data available

Note: Design to EN 1992 based on cylinder strength, from EN relationship, required cube strength relates to a higher cylinder strength i.e. conservative in using cube strength for conformity

Similarly, design based on cube strength (e.g. BS 8110), EN tabulated values lead to lower cylinder strength than experimental data, i.e. conservative in evaluating existing designed based on cube strength ($f_{\text{cube}} = 50$ for $f_{\text{cyl}} = 40$, cf experimental value = 40+)

Relationship between Cylinder Strength and Cube Strength



CONCRETE SPECIFICATION

Determination of compressive strength – standard specimens

Cubes vs Cylinders

Although design equations in EN 1992-1 adopt standard cylinder compressive strength, testing cube specimens for conformity has the following advantages:

- (a) End preparation – none required for cubes, particularly for high strength concrete by grinding - costly and time consuming;**
- (b) Inadequately prepared ends of cylinders contribute to higher variability in resultant strength or unacceptable results;**
- (c) For the same size specimens, cubes require less concrete per specimen (reduction in resource and waste for disposal).**

Specimen size (100 mm vs 150 mm)

Although EN 206-1 specifies 150 mm size, SS 544-2 (BS 8500-2) permits the use of 100 mm size with the same requirements for conformity

Advantages of 100 mm cubes:

- (a) Volume of concrete used and less waste for disposal reduced by 2/3;**
- (b) Increased capacity of available curing facilities (3 times);**
- (c) Capacity of existing compression testing machine caters for 2.25 times higher strength (needed for testing high strength concrete).**

Specification of Concrete

Consistence classes – (in fresh concrete presentation)

Selection of test method(s) and corresponding class depends on nature of project (designer input) and suitable for conditions of handling and placing method (contractor input) to enable concrete after compaction to surround all reinforcement, tendons, and ducts and completely filled the formwork.

Selection of consistence: provided in BS 8500-2 (SS 544-2), Annex A (informative) Guidance for the specifier (Table A.16).

EN 206: 2013 Annex G (informative) Guidance for self-compacting concrete requirements in the fresh state

Further guidance on production and other aspects of SCC:

The European guidelines for self-compacting concrete – Specification – Production and use., May 2005

(<http://www.efnarc.org/pdf/SCCGuidelinesMay2005.pdf>)

Specification of Concrete

SS 544-1 2009 Table A.16 Consistence suitable for different uses of in-situ concrete

Table A.16 – Consistence suitable for different uses of in-situ concrete

Use of concrete	Form of compaction	Consistence class	
		Normal-weight concrete	Lightweight concrete
Kerb bedding and backing	Tamping	S1	–
Floors and hand placed pavements	Poker or beam vibration	S2	S2
Strip footings	Poker or beam vibration and/or tamping	S3	–
Mass concrete foundations		S3	–
Blinding		S3	–
Normal reinforced concrete in slabs, beams, walls and columns		S3	S3
Sliding formwork construction		S2	S2
Pumped concrete		S3	F5
Vacuum processed concrete		S3	S3
Trench fill	Self-weight compaction	S4	–
In-situ piling		S4	–

EXPOSURE CLASSES

BS EN 206-1 Exposure classes related to environmental actions

Table 1 – Exposure classes (in descriptive but not quantitative terms)

- 1. No risk of corrosion or attack : X0**
- 2. Corrosion induced by carbonation : XC1, XC2, XC3, XC4**
- 3. Corrosion induced by chlorides other than from sea water : XD1, XD2, XD3**
- 4. Corrosion induced by chlorides from sea water : XS1, XS2, XS3**
- 5. Freeze/thaw attack with or without de-icing agents : XF1, XF2, XF3, XF4**
- 6. Chemical attack : XA1, XA2, XA3**

(Table 2 – limiting values for exposure classes for chemical attack from natural soil and ground water)

BS EN 206 Annex F (informative), provides guidance on concrete to resist exposure classes in Table F.1 Recommended limiting values for composition and properties of concrete

(intended working life of 50 years, CEM I strength class 32.5, aggregate with maximum nominal upper size in the range of 20 mm to 32 mm)

BS 8500-1 Annex A (informative) UK recommendations which combines concrete cover together with specified limiting values for composition of properties of concrete for intended working life of 50 and 100 years. Certification of concrete for exposure class to EN 206 is based on a single set of limiting values for each exposure class

Comment: Details on design for durability under topic of Durability

Specification of Concrete

Table F.1 EN 206: 2013 (Table F.1 SS EN 206-1: 2009, updated)

Table F.1 — Recommended limiting values for composition and properties of concrete

	Exposure classes																	
	No risk of corrosion or attack	Carbonation-induced corrosion				Chloride-induced corrosion						Freeze/thaw attack				Aggressive chemical environments		
						Sea water			Chloride other than from sea water									
		X0	XC 1	XC 2	XC 3	XC 4	XS 1	XS 2	XS 3	XD 1	XD 2	XD 3	XF 1	XF 2	XF 3	XF 4	XA 1	XA 2
Maximum w/c^a	—	0,65	0,60	0,55	0,50	0,50	0,45	0,45	0,55	0,55	0,45	0,55	0,55	0,50	0,45	0,55	0,50	0,45
Minimum strength class	C12/15	C20/25	C25/30	C30/37	C30/37	C30/37	C35/45	C35/45	C30/37	C30/37	C35/45	C30/37	C25/30	C30/37	C30/37	C30/37	C30/37	C35/45
Minimum cement content ^c (kg/m ³)	—	260	280	280	300	300	320	340	300	300	320	300	300	320	340	300	320	360
Minimum air content (%)	—	—	—	—	—	—	—	—	—	—	—	—	4,0 ^a	4,0 ^a	4,0 ^a	—	—	—
Other requirements	—	—	—	—	—	—	—	—	—	—	—	Aggregate in accordance with EN 12620 with sufficient freeze/thaw resistance				—	Sulfate-resisting cement ^b	

^a Where the concrete is not air entrained, the performance of concrete should be tested according to an appropriate test method in comparison with a concrete for which freeze/thaw resistance for the relevant exposure class is proven.

^b Where sulfate in the environment leads to exposure classes XA2 and XA3, it is essential to use sulfate-resisting cement conforming to EN 197-1 or complementary national standards.

^c Where the k -value concept is applied the maximum w/c ratio and the minimum cement content are modified in accordance with 5.2.5.2.

SS 544-1 (BS 8500-1) Annex A provides more specific guidance (Topic including presented under Durability)

EXPOSURE CLASSES

SS 544-1: 2009 adopts UK guidance for chemical attack, BS 8500-1

Annex A (informative) plus additional guidelines

Table A.2 – Classification of ground conditions

Clause A.4.2 Concrete properties and limiting values to resist corrosion of reinforcement

Table A.4 Durability recommendations for reinforced or prestressed elements with an intended working life of at least 50 years

Table A.5 Durability recommendations for reinforced or prestressed elements with an intended working life of at least 100 years

The provisions are based on combinations of minimum cement content and maximum water/cement ratio and nominal cover (allowance for deviation, Δc added to the minimum cover to give nominal cover)

Tables A.9, A.10 and A.11 for selecting requirements to meet chemical attack

**Aggressive chemical environment for concrete, ACEC-class
(based on BRE Special Digest 1, 2005)**

For chemical attack, the guidance is different between EN 206 and BS 8500-1 – based on BRE Special Digest 1, 2005)

Conformity Control for Designed Concrete

**(Based on SS EN 206-1: 2009, and)
(SS 544-1: 2009 and SS 544-2: 2009)**

**Note: Table numbering may differ after updating to EN
206: 2013 and BS 8500-1: 2006 + A1: 2012
and BS 8500-2: 2006 + A1: 2012**

**There may also be changes in requirements and
addition of new requirements**

Conformity Control and Conformity Criteria

SS EN 206-1 Clause 8.2 Conformity control for designed concrete
Conformity control for compressive strength

Sampling and testing shall be performed either on individual concrete compositions or on concrete families of established suitability (see 3.1.14) as determined by the producer unless agreed otherwise

The family concept shall not be applied to concrete with higher strength class (C55/67 or LC55/60)

In the sampling and testing plan and the conformity criteria of individual concrete composition or concrete families, distinction is made between initial production and continuous production

Initial production covers the production until at least 35 test results are available.

Continuous production is achieved when at least 35 test results are obtained over a period not exceeding 12 months.

If the production of an individual concrete composition, or a concrete family, has been suspended more than 12 months, the producer shall adopt the criteria, sampling and testing plan and the criteria for initial production.

During continuous production, the producer may adopt the sampling plan and testing plan and the criteria for initial production

Conformity Control and Conformity Criteria

Sampling and Testing Plan

Sampling of concrete shall be randomly selected

Sampling shall be carried out on each family of concrete

The minimum rate of sampling and testing of concrete shall be in accordance with Table 13 at the rate that gives the highest number of samples for initial and continuous production.

Table 13 - Minimum rate of sampling for assessing conformity

Production	Minimum rate of sampling		
	First 50 m ³ of production	Subsequent to first 50 m ³ of production ^a	
		Concrete with production control certification	Concrete without production control certification
Initial (until at least 35 test results are obtained)	3 samples	1/200 m ³ or 2/production week	1/150 m ³ or 1/production week
Continuous ^b (when at least 35 test results are available)		1/400 m ³ or 1/production week	

^a Sampling shall be distributed throughout the production and should not be more than 1 sample within each 25 m³.

^b Where the standard deviation of the last 15 test results exceeds 1.37σ , the sampling rate shall be increased to that required for initial production for the next 35 test results.

Conformity Control and Conformity Criteria

The samples shall be taken after any water or admixtures are added to the concrete under the responsibility of the producer, but sampling before adding plasticizer or superplasticizer to adjust consistence is permitted where there is proof by initial testing that the plasticizer or superplasticizer in the quantity to be used has no negative effect on the strength of concrete.

The test result shall be that obtained from an individual specimen or the average of the results when two or more specimens made from one sample are tested at the same age.

Where two or more specimens are made from one sample and the range of the test values is more than 15% of the mean then the results shall be disregarded unless an investigation reveals an acceptable reason to justify disregarding an individual test value

Comment:

Single test specimen is risky (workmanship, handling, and testing)

More than two has higher probability of range exceeding 15% of mean)

e.g. range of 3 specimens may be up to 20%

SS 544-2 Clause 12 Conformity testing and conformity criteria

Conformity testing of ready-mixed concrete shall be based on samples taken at or before delivery.

Conformity Control and Conformity Criteria

Conformity criteria for compressive strength

Conformity assessment shall be made on test results during an assessment period that should not exceed the last twelve months.

Conformity of concrete compressive strength is assessed on specimens tested at 28 days (or other specified age) for:

- groups of “n” non-overlapping or overlapping consecutive test results f_{cm} (Criterion 1);
- each individual test result f_{ci} (Criterion 2).

NOTE – The conformity criteria are developed on the basis of non-overlapping test results. Application of the criteria to overlapping test results increases the risk of rejection

Conformity is confirmed if both the criteria given in Table 14 for either initial or continuous production are satisfied.

Table 14 – Conformity criteria for compressive strength

Production	Number n of test results for compressive strength in the group	Criterion 1	Criterion 2
		Mean of n results (f_{cm}) – N/mm	Any individual test result (f_{ci}) – N/mm ²
Initial	3	$\geq f_{ck} + 4$	$\geq f_{ck} - 4$
Continuous	Not less than 15	$\geq f_{ck} + 1.48 \sigma$	$\geq f_{ck} - 4$

Conformity Control and Conformity Criteria

Conformity criteria for compressive strength - continued

Where conformity is assessed on the basis of a concrete family, Criterion 1 is to be applied to the reference concrete taking into account all transposed test results of the family; Criterion 2 is to be applied to the original test results.

To conform that each individual member belongs to the family, the mean of non-transposed test results (f_{cm}) of a single family member shall be assessed against Criterion 3 as given in Table 15.

Any concrete failing this criterion shall be removed from the family and assessed individually for conformity.

Table 15 – Confirmation criterion for family member

Number n of test results for compressive strength for a single concrete	Criterion 3
	Mean of n results (f_{cm}) for a single family member N/mm ²
2	$\geq f_{ck} - 1.0$
3	$\geq f_{ck} + 1.0$
4	$\geq f_{ck} + 2.0$
5	$\geq f_{ck} + 2.5$
6	$\geq f_{ck} + 3.0$

Conformity Control for Designed Concrete

Conformity control for properties other than strength

Table 18 — Confirmation criterion for family members

Number n of test results for compressive strength for a single family member	Mean of n results (f_{cm}) for a single family member N/mm ²
2	$\geq f_{ck} - 1,0$
3	$\geq f_{ck} + 1,0$
4	$\geq f_{ck} + 2,0$
5	$\geq f_{ck} + 2,5$
6	$\geq f_{ck} + 3,0$
7 to 9	$\geq f_{ck} + 3,5$
10 to 12	$\geq f_{ck} + 4,0$
13, 14	$\geq f_{ck} + 4,5$
≥ 15	$\geq f_{ck} + 1,48 \sigma$

Table 18 extends number n of test results for a single family member in Table 15 of SS EN 206-1: 2009 to beyond 6

Conformity Control and Conformity Criteria

Conformity criteria for compressive strength - continued

Initially the standard deviation shall be calculated from at least 35 consecutive test results taken over a period exceeding three months and which is immediately prior to the production period during which conformity is to be checked.

This value shall be taken as the estimate of the standard deviation (σ) of the population.

The validity of the adopted value has to be verified during the subsequent production.

Two methods of verifying the estimate of the value of σ are permitted, the choice shall be made in advance.

Method 1

The initial value of standard deviation may be applied for the subsequent period during which conformity is to be checked, provided the standard deviation of the latest 15 results (s_{15}) does not deviate significantly from the adopted standard deviation.

This is considered valid provided:

- **$0.63 \sigma \leq s_{15} \leq \sigma 1.37$**
- Where the value of s_{15} lies outside these limits, a new estimate of σ shall be determined from the last available 35 test results.

Conformity Control and Conformity Criteria

Conformity criteria for compressive strength - continued

Method 2

The new value of σ may be estimated from a continuous system and this value is adopted. The sensitivity of the system shall be at least shall be at least that of Method 1.

The new estimate for σ or the value of s_{15} lies outside these limits, a new estimate of σ shall be applied to the next assessment period.

Guidance on the application of the EN 206-1 conformity rules Quarry Products Association, April 2004

Estimation of the standard deviation based upon:

Estimated standard deviation = $0.886 \times \text{mean range of successive pairs of results}$

Example: Range, R = difference between two consecutive test results
15 sets of cube strength providing 14 values for the range

Estimated standard deviation = $0.886 \times \Sigma R_{14} / 14$

The use of R is a feature associated with the CUSUM system for quality control commonly adopted in UK ready-mixed concrete industry.

This approach is in EN 206:2013, Annex H (informative) as Method C –
Use of control charts

Conformity Control and Conformity Criteria

SS EN 206-1 Compressive strength class (Clause 4.3.1)

The characteristic compressive strength at 28 days of 150 mm diameter by 300 mm cylinders ($f_{ck,cyl}$) or the characteristic compressive strength at 28 days of 150 mm cubes ($f_{ck,cube}$) may be used for classification

SS EN 206-1 Requirements for hardened concrete (Clause 5)

Where the strength is to be determined, it shall be based on tests carried out on either 150 mm cubes or 150/300 mm cylinders conforming to EN 12390-1 and made and cured in accordance with EN 12390-2 from samples taken in accordance with EN 12350-1.

SS 544-2 Conformity control for compressive strength (Clause 12.2)

If conformity to the specified compressive strength class is determined using 100 mm cubes, the minimum characteristic 100 mm cube strength shall be that given for 150 mm cubes in SS EN 206-1: 2009, Table 7 and SS EN 206-1: 2009 Table 8.

(HDB has adopted use of 100 mm cubes)

BS 12504-1: 2009 Taking cores (Clause 5)

The ratio of the maximum aggregate size in the concrete to the diameter of the core has a significant influence on the measured strength when it approaches values greater than about 1:3.

Conformity Control and Conformity Criteria

SS EN 206-1 Definition (Clause 3)

3.1.14 Concrete family

Group of concrete compositions for which a reliable relationship between relevant properties is established and documented

Annex K (informative) Concrete families

Where there is little experience of using the concrete family concept, the following is recommended for a family:

- cement of one type, strength class and source;
- demonstrably similar aggregates and type I additions;
- concretes with or without a water reducing/plasticizing admixture;
- full range of consistence classes;
- concretes with a limited range of strength classes.

Before using the family concept or extending the families given above, the relationships should be tested on previous production data to prove that they give adequate and effective production and conformity control

Conformity assessed on the basis of a concrete family has the additional requirement to confirm each member belongs to the family (see SS EN 206-1: 2009 Table 15 provided earlier)

Conformity Control and Conformity Criteria

SS EN 206-1 compared to SS 289 (BS 5328)

SS 289 (BS 5328)

Two compressive strength test specimens for each test age

Mean of two results shall be taken as the test result

When the difference between the two results divided by their mean exceeds 15%, the test result shall be deemed invalid

Group of concrete compositions for which a reliable relationship between relevant properties is established and documented

SS EN 206-1

The test result shall be that obtained from an individual specimen or the average of the results when two or more specimens made from one sample are tested at the same age.

Where two or more specimens are made from one sample and the range of the test values is more than 15% of the mean then the results shall be disregarded unless an investigation reveals an acceptable reason to justify disregarding an individual test value

Comment:

Recommend to retain 2 specimens for each test age, spare specimens, if available, may be kept for testing at later ages

SS EN 206-1 compared to SS 289 (BS 5328)

SS 289 (BS 5328)

For C20 and above, group of test result of any consecutive 4:

Mean of group of test results to exceed specified characteristic compressive strength by at least 3 MPa

For first 3 test results, by at least 2 MPa

Any individual test result is not less than characteristic compressive strength less 3 MPa (also for first 3 test results)

SS EN 206-1 (BS EN 206-1)

Table 14 – Conformity criteria for compressive strength

Production	Number n of test results for compressive strength in the group	Criterion 1	Criterion 2
		Mean of n results (f_{cm}) – N/mm ²	Any individual test result (f_{ci}) – N/mm ²
Initial	3	$\geq f_{ck} + 4$	$\geq f_{ck} - 4$
Continuous	Not less than 15	$\geq f_{ck} + 1.48 \sigma$	$\geq f_{ck} - 4$

Comment:

EN Criterion 1 for individual is lower,

EN Criterion 2 for mean is higher for initial production, and also for continuous production when $\sigma > 2$ MPa

Conformity Control for Designed Concrete

Similar approach is provided for conformity criteria for tensile splitting strength (SS EN 206-1, 8.2.2)

Conformity criteria for properties other than strength

SS EN 206-1, 8.2.3 Table 17 includes Density, water-cement ratio, cement content (based on as recorded on print-out of batch record or from production record in connection with batching instruction), air content of air-entrained fresh concrete and chloride content (based on sum of contributions from constituent materials)

Conformity criteria for consistence (SS EN 206-1, 8.2.3 Table 18) Example for slump is used to illustrate the approach. The same approach is adopted for properties in Table 17 and Table 18

- ★ Individual test results are within maximum allowed deviation in Tables 17 and 18**
- ★ Number of test results outside specified limiting value, class limits or tolerances of target values as appropriate is not greater than the acceptance number in Tables 19a or 19b as given in Tables 17 and 18, alternatively, requirement may be based on testing by variables in accordance with ISO 3951: 1989 Table II-A (AQL = 4%) where the acceptance number relates to Table 19a**

Conformity for Consistence

Example - BS EN 206-1 Table 3 'Slump classes'

Clause 4.2.1 Consistence classes (formerly workability)

Table 3 – Slump classes

Class	Slump in mm
S1	10 to 40
S2	50 to 90
S3	100 to 150
S4	160 to 210
S5 ¹⁾	≥ 220

¹⁾ Note to 5.4.1 Due to the lack of sensitivity of the test methods beyond certain values of consistence, it is recommended to use the indicated tests for:

slump ≥ 10 mm and ≤ 210 mm (Slump test conforming to EN 12350-2)

In special cases, consistence may also be specified by target value

Table 11 – Tolerances for target values of consistence

Slump			
Target value in mm	≤ 40	50 to 90	≥ 100
Tolerance in mm	± 10	± 20	± 30

Conformity for Consistence

Example – BS EN 206-1 Table 3 ‘Slump classes’ (cont’d)
BS EN 206-1 **Table 18** ‘Conformity criteria for consistence

Test method		Maximum number of samples or determinations	Acceptance number	Maximum allowed deviation of single test results from the limits of the specified class or from the tolerance on the target value	
				Lower value	Upper value
Slump	EN 12350-2	Frequency as for compressive strength	see Table 19 b (AQL = 15%)	– 10 mm	+ 20 mm
				– 20 mm*	+ 30 mm*
* Only applicable for consistence testing from initial discharge from truck mixer					

BS EN 206-1 Table 19b ‘Acceptance number for conformity criteria for properties other than strength’

Number of test results	1 - 2	3 - 4	5 - 7	8 - 12	13 - 19
Acceptance number	0	1	2	3	5
Number of test results	20 – 31	32 - 49	50 - 79	80 - 100	
Acceptance number	7	10	14	21	

For 8 to 12 samples taken on same day: no slump outside 80 to 180 mm and not more 3 out range of 100 to 150 mm

Conformity Control for Designed Concrete

Conformity control for properties other than strength

EN 206: 2013 Tables 21, 22, 23 and 24 replaced

SS EN 206-1: 2009 Tables 17, 18, 19a and 19b

Conformity with required property is confirmed where both

- ♦ **Individual test results are within the maximum allowed deviation given in Tables 21 and 22 or the tolerances on target values conform to Table 23; and**
- ♦ **the number of test results for the property given in Table 22 outside the specified limiting value or class limits or tolerance on target values as appropriate is not greater than the acceptance number in Table 24; (alternatively, the requirement may be based on testing by variables in accordance with ISO 3951-1 (AQL = 4%) – [AQL is Acceptance Quality Limit]**
- ♦ **Assessment period shall not exceed 6 months (changed from 12 months in SS EN 206-1: 2009)**

Conformity Control for Designed Concrete

Conformity control for properties other than strength

Table 21 — Conformity assessment for consistence classes, SCC properties, air content and homogeneity of fibre distribution of fresh concrete at the point of delivery

Property	Test method or method of determination	Minimum number of samples or determinations	Maximum allowed deviation ^a at the point of delivery of single test results from limit values or for consistence the limits of the specified class	
			Lower limit	Upper limit
Appearance	Comparison by visual inspection of the appearance of this concrete with its normal appearance	Each batch; for vehicle deliveries, each load	-	-
Slump	EN 12350-2	i) Frequency as given in Table 17 for compressive strength ii) When testing air content iii) In case of doubt following visual inspection	-10 mm	+10 mm
Degree of compactibility	EN 12350-4		-20 mm ^b	+20 mm ^b
			-0,03	+0,03
			-0,04 ^b	+0,04 ^b
Flow	EN 12350-5		-10 mm	+10 mm
Slump flow	EN 12350-8		-20 mm ^b	+20 mm ^b
Viscosity	EN 12350-8 or EN 12350-9	If specified	No deviation allowed	No deviation allowed
Passing ability	EN 12350-10 or EN 12350-12			
Segregation resistance	EN 12350-11			
Air content of air-entrained fresh concrete ^d	EN 12350-7 for normal-weight and heavy-weight concrete and ASTM C 173 for lightweight concrete	1 sample / production day ^c	-0,5 % by volume	+5,0 % by volume
Homogeneous mixing of fibres in fresh concrete where the fibres are added at the truck mixer	As given in B.5	Frequency ^c as given in Table 17 for compressive strength	As given in B.5	

^a Where there is no lower or upper limit in the relevant consistence class, these deviations do not apply.

^b Only applicable for consistence testing from initial discharge from truck mixer or agitating equipment (see 5.4.1).

^c Except where provisions valid in the place of use require higher minimum test rates.

^d See 6.2.3 (1), fourth bullet point.

Conformity Control for Designed Concrete

Conformity control for properties other than strength - continued

Table 22 — Conformity assessment for fibre content, density, maximum water/cement ratio and minimum cement content

Property	Test method or method of determination	Minimum number of samples or determinations	Acceptance number	Maximum allowed deviation of single test results from limit values, from tolerance on a target value or from the limits of the specified class	
				Lower limit	Upper limit
Steel fibre content of fresh concrete	See 5.4.4	1 determination per day	See Table 24	-5 % by mass	No limit ^a
Polymer fibre content of fresh concrete	See 5.4.4	1 determination per day	See Table 24	-10 % by mass	No limit ^a
Density of heavy-weight concrete	EN 12390-7	As Table 17 for compressive strength	See Table 24	-30 kg/m ³	No limit ^a
Density of lightweight concrete	EN 12390-7	As Table 17 for compressive strength	See Table 24	-30 kg/m ³	+30 kg/m ³
Maximum Water/cement ratio, or Maximum Water/(cement + addition) ratio ^b , or Maximum Water/(cement + $k \times$ addition) ratio ^b	See 5.4.2	1 determination per day	See Table 24	No limit ^a	+0,02
Minimum cement content, or Minimum (cement + addition) content ^b , or Minimum (cement + $k \times$ addition) content ^b	See 5.4.2	1 determination per day	See Table 24	-10 kg/m ³	No limit ^a
^a Unless limits are specified. ^b Depending on the addition concept in use, see 5.4.2.					

Conformity Control for Designed Concrete

Conformity control for properties other than strength – continued

Table 23 — Conformity criteria for target values ^a of consistence and viscosity

Slump			
Target value in mm	≤ 40	50 to 90	≥ 100
Tolerance in mm	± 10	± 20	± 30
Degree of compactability			
Target value	≥ 1,26	1,25 to 1,11	≤ 1,10
Tolerance	± 0,13	± 0,11	± 0,08
Flow diameter			
Target value in mm	All values		
Tolerance in mm	± 40		
Slump flow diameter			
Target value in mm	All values		
Tolerance in mm	± 50		
t₅₀₀			
Target value in s	All values		
Tolerance in s	± 1		
t_v			
Target value in s	< 9	≥ 9	
Tolerance in s	± 3	± 5	

^a These values apply except where alternative values are given in Annex D or in provisions valid in the place of use.

Conformity Control for Designed Concrete

Conformity control for properties other than strength - continued

Table 24 — Acceptance numbers for conformity criteria given in Table 22

AQL = 4 %	
Number of test results	Acceptance number
1 to 12	0
13 to 19	1
20 to 31	2
32 to 39	3
40 to 49	4
50 to 64	5
65 to 79	6
80 to 94	7
95 to 100	8
Where the number of test results exceeds 100, the appropriate acceptance numbers may be taken from ISO 2859-1:1999, Table 2-A.	

Identity Test

With structural concrete being produced under certification scheme, the need to carry out regular sampling on site in current practice may be reduced – required by BCA since 2011

The production control test results for a plant form a much larger population as they are taken by the producer for the same designed concrete (including family concept, if adopted) compared to population based on a single site.

When the need to verify if a batch delivered to site complies with the specification, additional sampling is specified under “identity testing”

Comment

Current practice is equivalent to sampling up to 100% of concrete delivered to site for identity testing

SS EN 206-1: 2009 Annex B (normative), Identity testing for compressive strength

Concrete under production control certification (B.3.1)

Concrete not under production control certification (B.3.2):

Identity Testing for Compressive Strength

IDENTITY TESTING FOR COMPRESSIVE STRENGTH

BS EN 206-1 Clause 8.2.1.1

Where identity of a defined volume of concrete with a population verified as conforming to the characteristic strength requirements is to be assessed, e.g. if there is doubt about the quality of a batch or load or if in special cases required by the project specification, this shall be in accordance with Annex B.

Annex B (normative) Identity Testing for Compressive Strength

SS 544-1: 2009 Annex B (normative) Identity testing for slump, flow, air content, density and additional requirements for compressive strength

(Identity criteria for other than compressive strength – refer to Annex B as these are not provided under SS EN 206-1: 2009)

Identity Testing for Compressive Strength

BS EN 206-1 Annex B (normative)

Identity testing indicates whether the defined volume of concrete in question belongs to the same population as that verified as conforming with the characteristic strength via conformity assessment by the producer.

Particular volume of concrete shall be identified, e.g.:

- single batch or load where there is doubt as to the quality;**
- concrete supplied to each storey of a building or group of beams/slabs or column/wall of a storey, or comparable parts of other structures;**
- concrete delivered to a site within one week, but not more than 400m³**

Number of samples taken from a particular volume of concrete shall be defined, and taken from different batches or loads

Average of the results of two or more specimens made from one sample for testing at the same age;

Range of the test results $\leq 15\%$ of the mean

Identity Testing for Compressive Strength

SS EN 206-1 Annex B (normative)

Concrete under production control certification

Table B.1 – Identity criteria for compressive strength

Number (n) of test results for compressive strength from the defined volume of concrete	Criterion 1	Criterion 2
	Mean of n results (f_{ck}) N/mm ²	Any individual test result (f_{ci}) N/mm ²
1	Not applicable	$\geq f_{ck} - 4$
2 to 4	$\geq f_{ck} + 1$	$\geq f_{ck} - 4$
5 to 6	$\geq f_{ck} + 2$	$\geq f_{ck} - 4$

For concrete NOT under production control certification:

From the defined volume of concrete, at least 3 samples shall be taken for testing

Conformity criteria as for initial production in Table 14:

Mean of 3 results (f_{cm}) $\geq (f_{ck} + 4)$ N/mm²

Any individual test result (f_{ci}) $\geq (f_{ck} - 4)$ N/mm²

Comment:

Number of samples may be related to total volume concrete defined

Identity Testing for Compressive Strength

SS 544-1 Annex B (normative)

B.5 Additional requirements for compressive strength

Identity testing shall be based on composite samples as defined in BS EN 12350-1

(Composite sample: quantity of concrete consisting of a number of increments distributed through a batch or mass of concrete, thoroughly mixed together)

Comment

Not commonly practised, usually sample taken from initial discharge

Each concrete shall be assessed individually

Within each defined volume of concrete, the individual results shall first be assessed against criterion 2 in SS 206-1: 2009 Annex B, Table B.1

Any result that appears to be an outlier shall be investigated to determine whether it is a valid result

Invalid results shall not be included in the calculation of the mean strength

Identity Testing for Compressive Strength

SS 544-1 Annex B (normative)

B.5 Additional requirements for compressive strength

The mean strength of all valid results in the volume of concrete shall be calculated and the result compared with SS EN 206-1: 2009, Annex B, Table B.1, criterion 1 if there are two to six results in the volume of concrete.

Where there are more than six results in the volume of concrete, the results shall be split into non-overlapping groups of six results taken in chronological sequence

SS EN 206-1: 2009, Annex B, Table B.1, criterion 1 shall be applied to each of the group of six results and, where applicable, the last group with less than six results

Comment

Identity testing is intended to reduce regular sampling for testing on site in current practice particularly with concrete certified by third party (Certification Body) accredited by SAC.

Identity testing incurs additional cost – prior agreement between user and producer to form part of supply contract

Example for Identity of Consistence

Example - SS EN 206-1 Table 3 'Slump classes'

Clause 4.2.1 Consistence classes (formerly workability)

Table 3 – Slump classes

Class	Slump in mm
S1	10 to 40
S2	50 to 90
S3	100 to 150
S4	160 to 210
S5 ¹⁾	≥ 220

¹⁾ Note to 5.4.1 Due to the lack of sensitivity of the test methods beyond certain values of consistence, it is recommended to use the indicated tests for:

slump ≥ 10 mm and ≤ 210 mm (Slump test conforming to EN 12350-2)

In special cases, consistence may also be specified by target value

Table 11 – Tolerances for target values of consistence

Slump			
Target value in mm	≤ 40	50 to 90	≥ 100
Tolerance in mm	± 10	± 20	± 30

Identity Testing for Slump

SS EN 544-1: 2009 (BS 8500-1:2006) -Tables B.1 and B.2

Table B.1 – Identity criteria for slump specified as a slump class

Dimensions in millimetres

Specified slump class	Requirement			
	For composite samples taken in accordance with BS EN 12350-1		For spot samples taken from initial discharge	
	Not less than	Not more than	Not less than	Not more than
S1	0	60	0	70
S2	40	110	30	120
S3	90	170	80	180
S4	150	230	140	240
S5	210	–	200	–

Table B.2 – Identity criteria for slump specified as a target value

Dimensions in millimetres

Specified target slump	Tolerance	
	For composite samples taken in accordance with BS EN 12350-1	For spot samples taken from initial discharge
≤40	–20, +30	–30, +40
50 to 90	–30, +40	–40, +50
≥100	–40, +50	–50, +60

Identity Testing for Slump

Example:

(a) Slump class: S3 (100 to 150 mm)

Identity test acceptance:

Based on spot samples taken from at initial discharge:

Slump values ≥ 80 mm and ≤ 180 mm

(b) Slump target value: 120 mm

Based on spot samples taken from at initial discharge:

Slump values $\geq (120 - 50) = 70$ mm and $\leq (120 + 60) = 180$ mm

Comment:

Similar range of slump values for acceptance as target value is approximately in the mid-range of S3 (slump to nearest 10 mm)